Task 1: Problem Definition and Scope

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1. Objective

Define the problem, scope, users, objectives, constraints, and measurable success criteria for the Puppet-Provisioned Delivery Promise Calculator.

2. Problem Statement

The project is a web application for managing delivery records and calculating promised delivery dates. The larger DevOps objective is to make the application buildable, testable, deployable, configurable, and recoverable through automation.

3. Target Users

• Customer: view and search delivery records.

• Operations staff: manage operational delivery status.

• Administrator: manage application data and access.

• Development/DevOps team: maintains source, CI/CD, containers and configuration.

4. MVP Scope

ID	Feature
F01	User login / role context
F02	Role-based access
F03	Create delivery
F04	View delivery records
F05	Update delivery
F06	Search by customer/product
F07	Promised-date calculation
F08	Status display
F09	Authorized status update
F10	Delivery details
F11	Dashboard counts
F12	Filtering/search
F13	Validation
F14	Error handling
F15	Health/verification support

5. Constraints

• Java 21 and Spring Boot with Maven.

• SQLite is used for persistence.

• Jenkins uses port 8080; the application uses port 8765.

• Docker uses host port 8766 mapped to container port 8765.

• The workflow must run on the Windows development machine.

6. Success Criteria

• Maven build succeeds.

• Selenium tests pass.

• Jenkins executes CI/CD.

• Docker image/container runs successfully.

• Puppet establishes and restores desired configuration.

• Browser can access the deployed application.

7. Result

The project scope was defined as a delivery-management application combined with a complete DevOps automation workflow.

Task 1 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.

Figure 1. Project GitHub repository and README showing the Delivery Promise Calculator project scope.

